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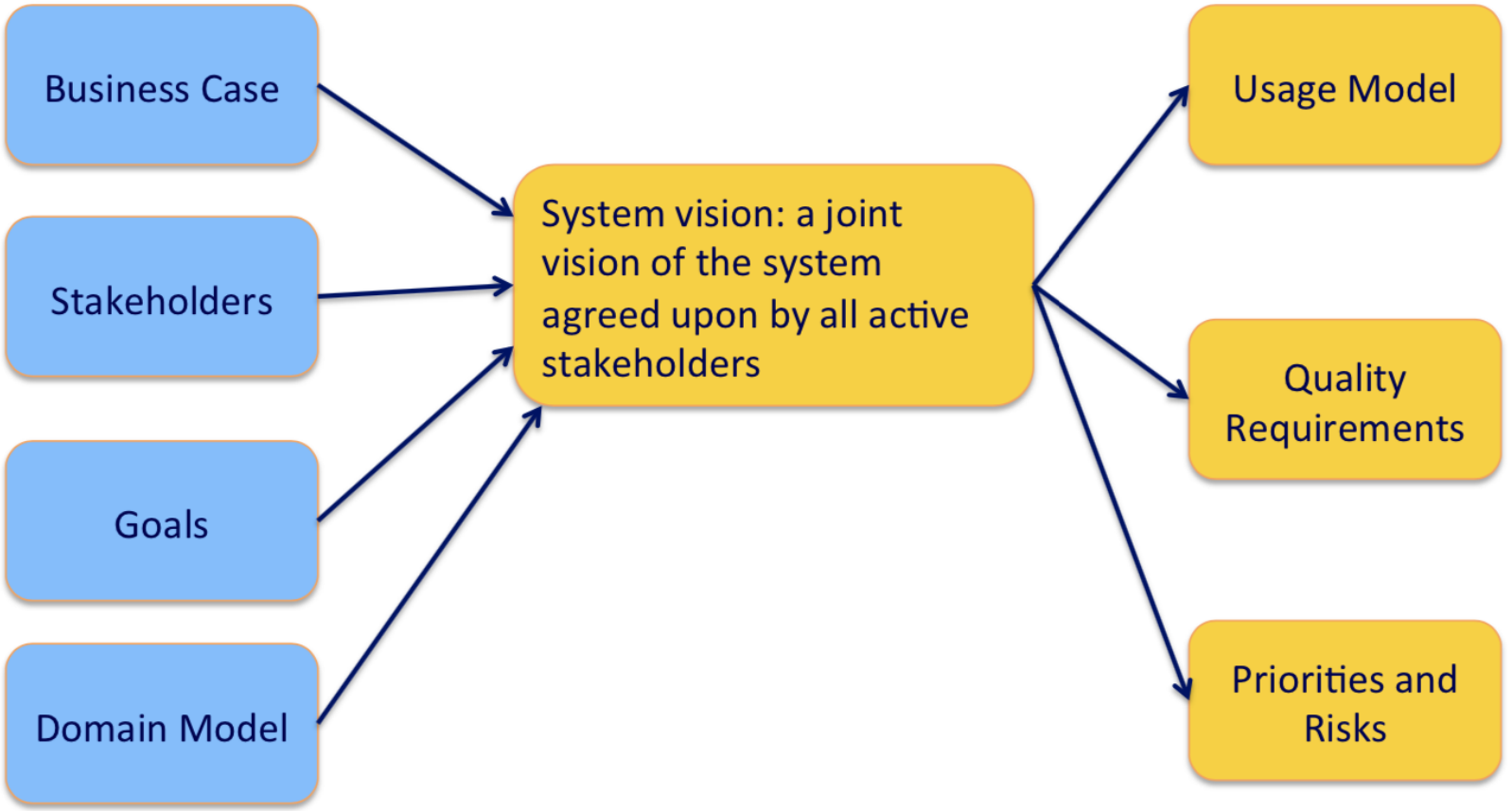
SE2203
Software Requirements Engineering

Lecture 4:
Usage Model. Use Cases

Recap time

The **vision and scope document** collects all business requirements into a single deliverable that sets the stage for subsequent development work:

It defines the business case for the project in nontechnical language.



Learning outcomes for Lecture 4

- What is a Usage Model?
- What are Use Cases?
- How do we specify/model Use Cases?
- What are scenarios?

User Requirements

- User Requirements
  - Describe what the user will be able to do with the product (goals or tasks to perform)
  - Allow to answer the question “What do you need to do with the system?”
  - **Example**: As a user, I need to save the calculation results as a pdf report.
  - Why you should not ask questions like “What do you want?” or “What do you want the system to do?”
- User goals
  - Must align with the higher-level business goals captured in the business requirements
  - Can identify:
    - Tasks that the user can perform
    - Use cases
    - Importance of non-functional requirements
    - Design and implementation constraints
    - Assumptions
    - User acceptance criteria

Example: Flight reservation system
What to ask?
What use cases you might identify?

Usage Model

- A usage model describes the system **behavior** from the user's point of view
- It is a "black box" view on the system
- Focus on modeling of interaction sequences – use cases
- **Techniques**:
  - Templates (structured text)
  - Use case overview diagram (top-level diagram)
  - UML activity diagram – used to represent system-internal interactions
  - Sequence Diagrams – usually used to represent interaction with user
    - UML Sequence Diagram
    - Message Sequence Charts (MSC)

Use Case

- A use case is a sequence of system events (incl. interactions between the system and one or more external actors) that are
  - triggered by an actor
    - An actor is a person, organization, or external system/application/device that plays a role in one or more interactions with the system
  - leads to results for the actor
- Use case modeling – one of the most popular semi-formal modeling technique currently used in requirements development
- To identify use cases, identify a specific objective the user may have: Listen for the phrase “I need to ….”

Use Case Specification (1)

- Use case specification consists of:
  - Unique identifier
  - Use case name
    - Should indicate the value the use case will deliver to the actor
    - Always begins with a verb
    - Also contains an object (a noun)
    - Optional: adjectives and adverbs
  - Actors
  - Brief textual description
  - Events that trigger the use case execution

- Often also includes:
  - Normal flow of events or default sequence between the actor and system
  - Preconditions and post-conditions for the use case
  - Alternative flows and exceptions
  - Business rules that affect the use case
- **Examples of use case names:**
  - Register for payroll deduction; order menu item from cafeteria; check flight status; check borrowing power

Use Case Specification (2)

- Use cases describe user requirements at a fairly high level of abstraction
- You may begin with an initial high-level analysis of the required use cases, which is more abstract and then refine each use case further as it comes into scope for each iteration/increment
- Moderate/large-size projects might require combining use cases into a more general use case.
  - Why ??
- **Example**:
  - Use cases
  - “Order a meal for lunch from cafeteria” and “order a meal for lunch from cafeteria”
  - Can be generalized to “order a meal for lunch”

Creating Use Cases – Method 1

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identify actors

- Start by identifying as many actors as possible.

how actors interact with system

- Ask how the actors interact with the system to identify an initial set of use cases

- On the diagram, connect the actors with the use cases with which they are involved.

make connections

- If an actor supplies information, initiates the use case, or receives any information as a result of the use case, then there should be an association between them

lines are undirected

- Generally, don't include arrowheads on the association lines due to confusion with information flow.

Creating Use Cases – Method 2

- Identify one actor
- Identify all the uses cases involving this actor
- Identify another actor
- Identify all the uses cases involving this actor
- Both approaches work
- Simplicity is the key – keep it simple as possible
- Goal: is to collect (just) enough information to understand the use case
- Initial version might be incomplete -- refinement
- In parallel to creating the diagram, write a very brief description of each use case

To identify actors and use cases...

- Who uses the system?
- What other systems use this system?
- Who gets information from this system?
- Who provides information to the system?
- Where in the company is the system used?
- Who supports and maintains the system?
- What are users in this role trying to accomplish?
- To fulfill this role, what do users need to be able to do?
- What are the main tasks of users in this role?
- What information do users in this role need to examine, create, or change?
- What do users in this role need to be informed of by the system?
- What do users in this role need to inform the system about?

What to identify?

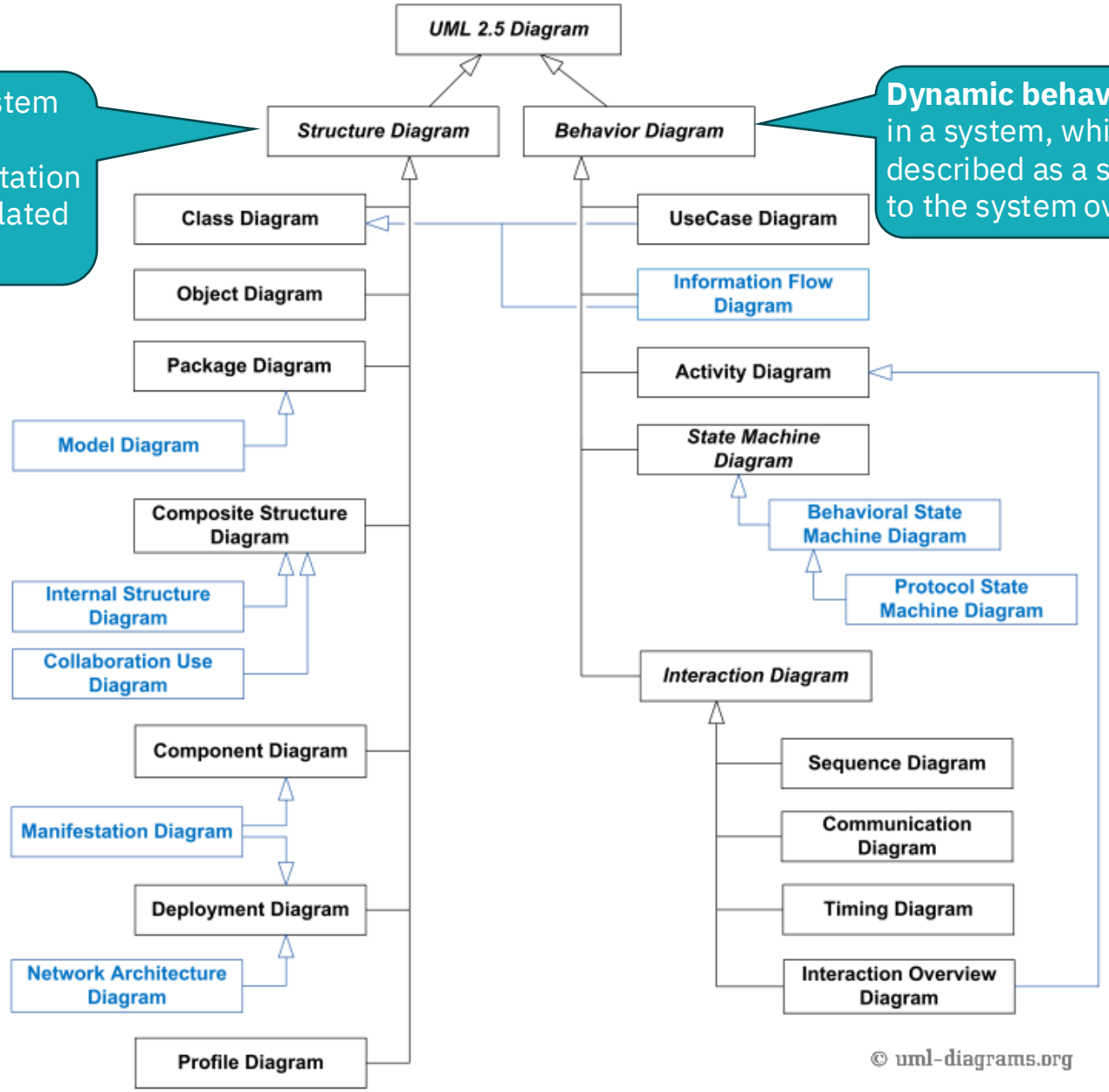
- Triggers that cause the scenario to begin
  - External events
  - Temporal conditions
- Inputs and outputs to the process
- Individual steps/actions that are taken
  - Show sequential order
  - Show conditional steps
- For each actor:
  - What will the actor use the system for?
  - Will the actor create, store, change, remove, or read data in the system?
  - Will the actor need to inform the system about external events or changes?
  - Will the actor need to be informed about certain occurrences in the system?

Example: Student Enrolment System

- In the example, **students** are enrolling in courses with the potential help of **registrars**. **Professors** input the marks students earn on assignments, and registrars authorize the distribution of transcripts (report cards) to students.

    - Who are the actors?

Static structure of the system and its parts on different abstraction and implementation levels and how they are related to each other



Dynamic behavior of the objects in a system, which can be described as a series of changes to the system over time

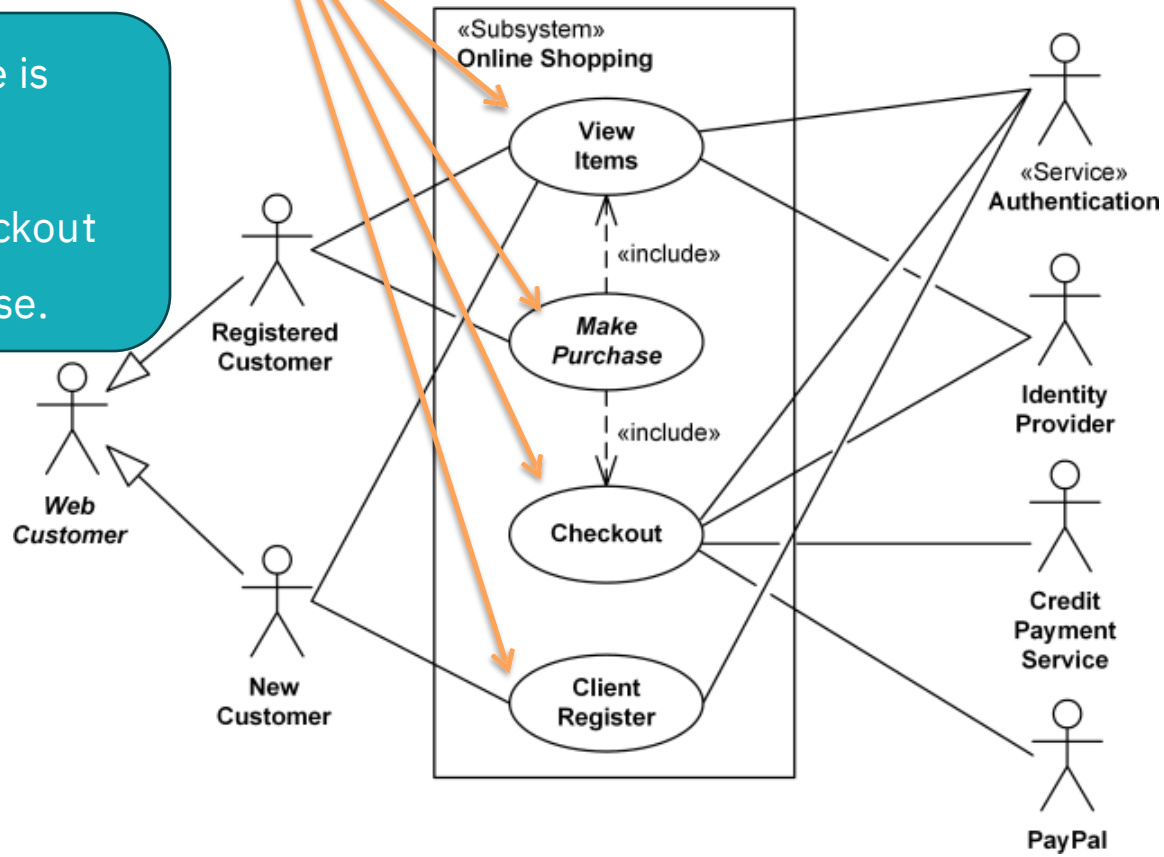
Unified Modelling Language (UML) is a general-purpose visual modelling language that is intended to provide a standard way to visualize the design of a system.

Example: Online shopping (Use Case Overview Diagram)

Top-level use cases

What use case is missing?

Note: Checkout use case is included use case not available by itself -- checkout is part of making purchase.

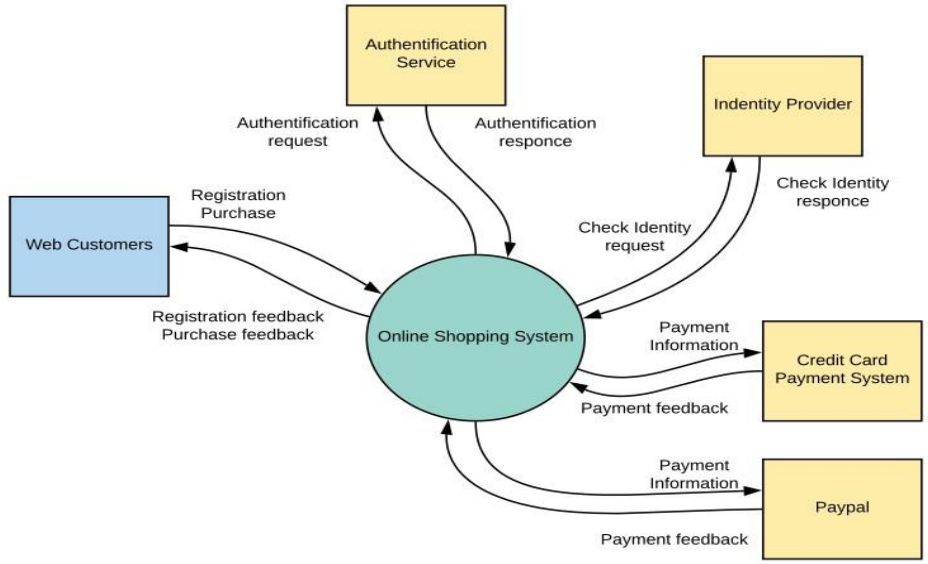


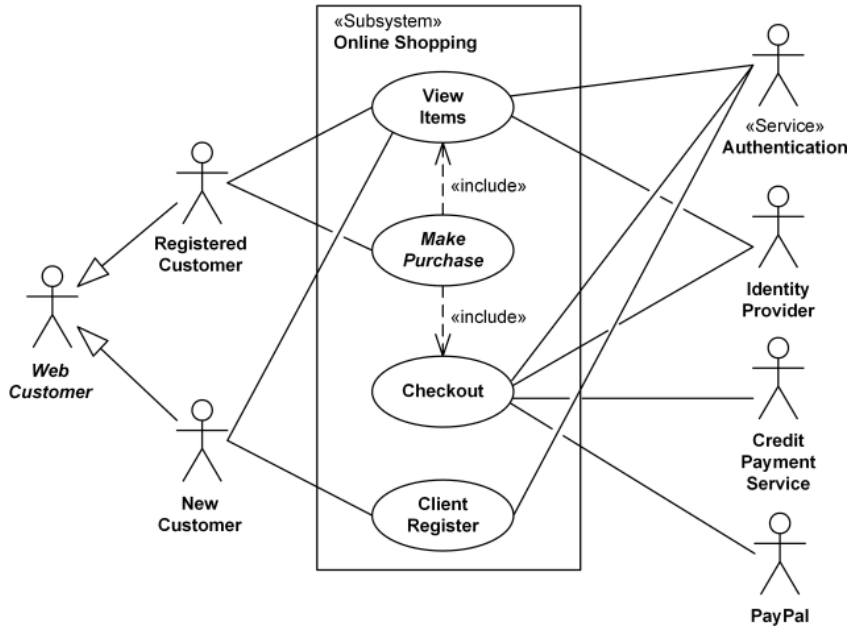
Note: Web Customer is not the only actor!

https://www.uml--diagrams.org/examples/online--shopping--use--case--diagram--example.html

Use Case vs. Context Diagram

- **Use case diagram**: the interaction of the system with its users
- **Context diagram**: the system's relationship with its environment





Scenarios

- Use case:
  - Set of all scenarios related to a task, objective, …
- Scenarios are ordered sets of interactions between a system and a group of external actors
  - Concrete sequences of events (interactions, steps, …)
  - Represent a specialization of the user goal in the use case, or alternative pathways to reach the user goal
  - Can be defined as a specific path through a use case or a specific instance of execution of a use case
- Some scenarios result in success, and the user achieves their intended goal
- Conditions under which they may fail to complete successfully called exceptions
- Most typical or default scenario is called the normal flow
- Other pathways through the use case that also result in success are alternative flows
- A use case can have zero, one or many alternative flows
- Functional requirement:
  - One/few event (step, interaction) from a scenario

Use Cases: Cockburn template

- Use cases with scenarios
- Structured text and/or

sequence interaction diagram

USE CASE #	< the name is the goal as a short active verb phrase>	
Goal in Context	<a longer statement of the goal in context if needed>	
Scope & Level	<what system is being considered black box under design> <one of : Summary, Primary Task, Subfunction>	
Preconditions	<what we expect is already the state of the world>	
Success End Condition	<the state of the world upon successful completion>	
Failed End Condition	<the state of the world if goal abandoned>	
Primary, Secondary Actors	<a role name or description for the primary actor>. <other systems relied upon to accomplish use case>	
Trigger	<the action upon the system that starts the use case>	
DESCRIPTION	Step	Action
	1	<put here the steps of the scenario from trigger to goal delivery, and any cleanup after>
	2	<...>
	3	
EXTENSIONS	Step	Branching Action
	1a	<condition causing branching> : <action or name of sub.use case>
SUB-VARIATIONS		Branching Action
	1	<list of variations>

RELATED INFORMATION	<Use case name>
Priority:	<how critical to your system / organization>
Performance	<the amount of time this use case should take>
Frequency	<how often it is expected to happen>
Channels to actors	<e.g. interactive, static files, database, timeouts>
OPEN ISSUES	<list of issues awaiting decision affecting this use case >
Due Date	<date or release needed>
...any other management information...	<...as needed>
Superordinates	<optional, name of use case(s) that includes this one>
Subordinates	<optional, depending on tools, links to sub.use cases>

Example with the Cockburn template (1)

USE CASE 5	Log in	
Goal in Context	Log in to the Online Shopping system	
Scope Level	Online Shopping system Primary task	
Preconditions	User is on the web page of the Online Shopping system	
Success End Condition	User logged in	
Failed End Condition	User is not logged in	
Primary, Secondary Actors	User -	
Trigger	User clicks on the link "Log in" in the header of the web page	
DESCRIPTION	Step	Action
	1	The log in box pops up. The following fields are available: • User name • Password • Forgot password or user name? • Register
	2	User enters the user name and password. User click on the Login button.
	3	The popup box disappears. The page is refreshed to the post-login page.

Example with the Cockburn template (2)

EXTENSIONS	Step	Branching Action
	2a	The user enters a wrong user name (there is no registered user with this name in the system) : The user name field is highlighted red. A red message above it says "Incorrect user name. There is no registered user with this name in the system. Please enter a correct user name". The user name field is empty, so that the user can enter correct name.
	2b	The user enters a wrong password (the password does not match to the entered user name) : The password field is highlighted red. A red message above it says "Incorrect password. The password does not match to the entered user name. Please check the name and password, and log in again". The password field is empty, so that the user can enter correct name.
SUB-VARIATIONS		
	V2c	User clicks on "Forgot password or user name?" Refer to the Use Case 8 for more details.
	V2d	User clicks on "Register". Refer to the Use Case 9 for more details.

RELATED INFORMATION	Use Case 5: Log in
Priority:	Top priority
Due Date	Version 1.0 release

Use Cases: Check list (1)

- Use Case title
  - Is it an active-verb goal phrase that names the goal of the primary actor?
  - Can the system deliver that goal?
- Primary actor:
  - Does they have behavior?
  - Does they have a goal against the SuD that is a service promise of the SuD?
    - SuD = System under Development (System under Discussion)
- Preconditions
  - Are they mandatory, and can they be set in place by the SuD?
  - Is it true that they are never checked in the use case?
- Main success scenario normal flow
  - Does it have 3-9 steps?
  - Does it run from trigger to delivery of the success guarantee?
  - Does it permit the right variations in sequencing?

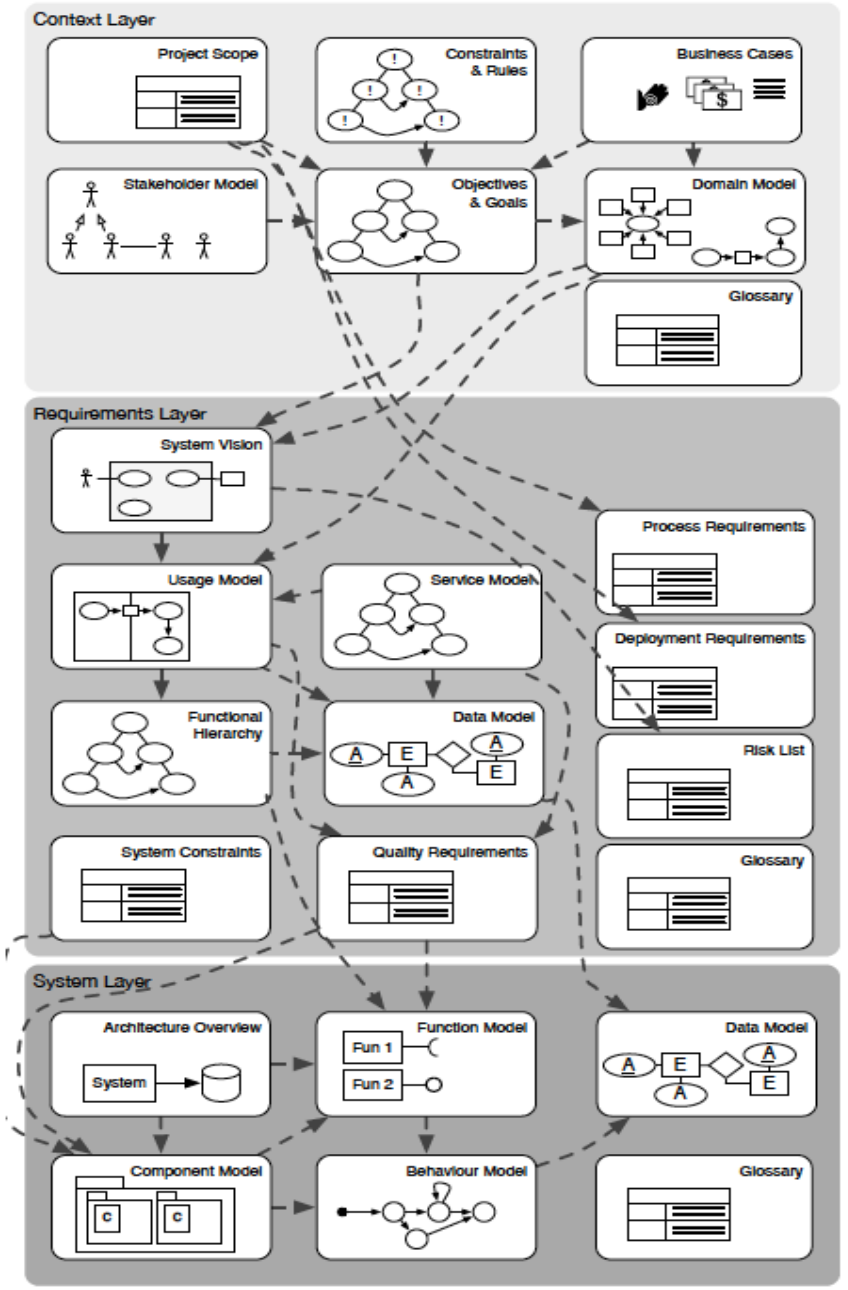
Use Cases: Check list (2)

- Each step in any scenario
  - Is it phrased as a goal that succeeds?
  - Does the process move distinctly forward after its successful completion?
  - Is it clear which actor is operating the goal – who is "kicking the ball”?
  - Is the intent of the actor clear?
  - Are you sure the step does not describe the user interface design of the system?
  - Is it clear what information is being passed in the step?
- Extension condition
  - Can and must the system both detect and handle it?
  - Is it what the system actually needs?
- Overall Use Case content
  - To the sponsors and users: "Is this what you want?”
  - To the sponsors and users: "Will you be able to tell, upon delivery, whether you got this?”
  - To the developers: "Can you implement this?”

AMDiRE Approach: Artifact-based RE

- Core parts of a usage model:
  - Use case overview diagram (“bubble” diagram)
  - Use case templates (one per “bubble”)
  - Scenario diagrams (one per use case template, i.e., one

Scenario diagrams (one per use case template, i.e., one per “bubble”)



Representation Techniques

What do we want to present?	What technique can we use?
System external interfaces	Context Diagram, Use Case Diagram Formal specifications, e.g., Focus[ST] Interface specifications (e.g, API)
Process	Activity Diagrams, Data Flow Diagram Flowcharts Formal specifications, e.g., Petri Nets
Relationships among data and/or objects	Entity-relationship diagram, Class Diagrams Data dictionary to define data structures and items
States within the system	State-transition diagram, Event-response tables Formal specifications
System behavior	State-transition diagram, State-chart diagram Formal specifications
Non-functional Requirements	Structured text, Planguage
User tasks	User stories, Use Case specifications, Sequence diagrams, Flowcharts, Activity Diagrams

4
4
3
3
3
2
5

no formal specification

Use Case vs. User Story

- Both identify users and they both describe goal, but…
- User Stories:
  - User role, goal and acceptance criteria
  - Deliberately leave out a lot of important details (provide a “bone” for eliciting during scrum meetings)
  - Centered on the result and the benefit of the thing you're describing (“As a registered user, I want to log in to the system”)
- Use Cases:
  - An actor, flow of events, and post-conditions, respectively etc.
  - More granular, and describe how your system will act – more detailed up-front requirement specification

Questions ?

- Essentially, all models are wrong, but some are useful. George E.P. Box

